

Karan Shah

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Work Experience

Adobe Inc.

San Jose, CA

MACHINE LEARNING ENGINEER

March 2021 - Current

- Working on GenAI advancements for Document Cloud products in the Document Cloud AI and Innovation Lab
- Building and upgrading ML pipelines for data generation-processing, model and performance evaluation, etc. for CV-NLP problems
- Working on ML training-inference engine deployments with ML Operational framework and tools (MLOps) to handle models in production

Quantiphi

Marlborough, MA

SOFTWARE DEVELOPER INTERN

Sept. 2020 - Dec. 2020

- Built software tools in Python and React.js for Machine Learning Engineers to accelerate production-grade Conversational AI use cases
- Leveraged Flask based routing methods with multiprocessing to trigger underlying capabilities of NVIDIA's end-to-end ML framework and Socket+GRPC for live audio transcription

Stony Brook University

Stony Brook, NY

GRADUATE TEACHING ASSISTANT, COMPUTER VISION

Jan. 2020 - May. 2020

- Assisted Prof. Dimitris Samaras in teaching Course CSE327-01: Computer Vision
- Managed assignment formulations and assessments, and resolve queries of students enrolled in the course

Simform

Ahmedabad, India

SOFTWARE ENGINEER

June 2018 - April 2019

- Developed an end to end Video Streaming system; allows user to upload and scan the content on the cloud, stream on cross-platform desktop application over globally separated clients; using AWS services (Lambda, S3, Cloudfront, MQTT), Electron.js, and React.js
- Built a software system to semi-automate Company's marketing work with Chrome Extension and REST API Web backend using Python and Node.js, efficient in reducing ~ 40 % of Marketing team's manual work
- Designed and automated ETL jobs to process a huge realtime data using AWS services (Glue, Lambda, S3) and Spark

ISRO - Indian Space Research Organization

Ahmedabad, India

RESEARCH INTERN, MACHINE LEARNING

July 2017 - May 2018

- Designed generalized CNN architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to Semantic Segmentation
- Built software tools in Python for preprocessing of geographic remote sensing data, visualization, real-time performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Skills

Programming Python, Javascript, C/C++, Matlab

Tools/Technologies REST, Git, Docker, Kubernetes, GRPC

Frameworks/Libraries PyTorch, Tensorflow, Keras, Node.js, Django, Flask, React.js, OpenCV

Education

Stony Brook University

Stony Brook, NY

M.S. IN COMPUTER SCIENCE; CONCENTRATION IN DATA SCIENCE AND ENGINEERING

Aug. 2019 - Dec. 2020

Coursework: Computer Vision, Machine Learning, Data Science, Probability and Statistics, Big Data Analytics, Theory of Database Systems

Gujarat Technological University

Ahmedabad, India

B.E. IN COMPUTER ENGINEERING

Aug. 2014 - May 2018

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Projects

Band-wise Independent Pansharpening Using CNN with Shared Weights (Python, Keras)

Jan. 2019 - July 2019

- Fine-tuned a Convolutional Neural Network to super-resolve Multispectral Satellite Imagery to Panchromatic Resolution
- Delivered high performance in terms of reduced resolution quantitative assessment and visual inspection
- Achieved better results compared to popular methods used for this purpose in the industry
- Submitted a **publication** with ISRO scientists to Geocarto International

Image Denoising using Deep Neural Networks (Python, Keras)

Oct. 2018 - Dec. 2018

- Designed a Deep Neural Network for Image Denoising and trained-performed on Overhead Images having Gaussian Noise
- Achieved computational efficiency and ability to generalize well despite limited training data
- Used Worldview Imagery samples provided by DigitalGlobe, achieving 90% accuracy

Video Action Classification - Recognition (Python, PyTorch, Matlab)

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved 86% (1st in class) on image test data, and 84% (2nd in class) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

Ultrasound Nerve Segmentation (Python, Tensorflow, Keras)

Nov. 2017 - Jan. 2018

- Designed and trained CNN based on U-net to segment a collection of Nerves in Ultrasound Images taken from a Kaggle competition
- Used pre-trained weights of VGG model in the Encoder of the network (transfer learning), gained 68 % Test Accuracy (Top 100)